

Dr. Dipankar Bhattacharya, Ph.D.

Marie Skłodowska-Curie Postdoctoral Fellow (2025–2027)

Dyson School of Design Engineering, Imperial College London, United Kingdom

d.bhattacharya1@imperial.ac.uk · dipankar22041988@gmail.com · +44 7448 647 225
<https://bhattacharya143.github.io/dips/> · LinkedIn · Room 302, Dyson Bldg, SW7 2DB, London

Summary Statement

Knowledge as such does not have much power unless it is transformed into technology, which is capable of accelerating the progress of mankind. The long cherished dream of becoming the orchestrate of transformation leads my eyes and heart towards the academics and research.

Education

- 2016 – 2021** **Ph.D., Mechatronics Engineering** in Department of Mechanical and Mechatronics Engineering, The University of Auckland (UoA), 01/08/2016 – 14/04/2021.
Thesis title: *Robotic Soft Esophagus Design, Modelling, and Control for Endoprosthetic Stent Testing and Food Swallow Investigation*
- 2011 – 2013** **M.Tech , Systems and Control** in Department of Electrical Engineering, Indian Institute of Technology (IIT), 22/07/2011 – 04/07/2013.
Dissertation title: *Non-Linear Predictive Control Model using Relevance Vector Machine*. GPA: **8.4 / 10.0**
- 2004 – 2010** **B.Tech , Electronics and Communication Engineering, North Eastern Regional Institute of Science and Technology (NERIST)**, 21/07/2004 – 23/06/2010.
Project title: *Design and Synthesis of a Sixth Order Tow Thomas Bi-quad filter*. GPA: **9.1 / 10.0**

Employment History

- 2025 – Present** **Marie Skłodowska-Curie Postdoctoral Fellow** in the Dyson School of Engg, Imperial College London, Supervisor: Prof. Thrishantha Nanayakkara (Morph Lab), 01/11/2025 – Present.
CASREx: Cable-driven Arm Soft Re-configurable Exoskeleton for Muscle Impairment Rehabilitation
- 2026 – Present** **Assistant Supervisor**, Dyson School of Design Engineering, Imperial College London, United Kingdom, 15/01/2026 – Present.
Supervising two Year 3 MEng (Design Engineering) placement students: **Ben Surtees** and **Lijiangke Liu** (March 2026 – Present). Research and development projects related to exosuit development.
Delivered lecture on **Robotics Simulation and Control using PyDrake** for **DESE71002 Robotics Research Projects**, Dyson School of Engineering, Imperial College London (28/01/2026).
- 2025** **Senior Research Engineer–Control Systems** in the Center for Transformative Garment Production (TransGP), Supervisor: Prof. Kazuhiro Kosuge, 01/05/2025 – 31/10/2025.
Physics-based Cloth Manipulation with policy generated from imitation learning.
- Led a multidisciplinary team to develop high-performance models for fabric pre-sewing handling and folding.

	• Investigated feasibility and oversaw development and validation of robotic systems for fabric manipulation. • Translated industrial requirements into research deliverables; contributed to papers, reports, and grant proposals. • Mentoring and supervising 3 PhD students; reviewing and proofreading manuscripts.
2024 – 2025	Postdoctoral Fellow in the Center for Transformative Garment Production (TransGP), Supervisor: Prof. Kazuhiro Kosuge, 02/07/2024 – 30/04/2025.
2024 – 2025	Visiting Research Associate in the Department of Electrical and Electronic Engineering, Hong Kong University (HKU), 02/07/2024 – 31/10/2025.
2021 – 2024	Postdoctoral Fellow in the Department of Mechanical and Automation, The Chinese University of Hong Kong. Supervisor: Prof. Darwin Lau, 01/03/2021 – 30/06/2024 <i>Project title 1: Modeling and control of cable-driven robots with cable wrapping around obstacles and links.</i> <i>Project title 2: Design of a cable-driven growing vine robot for safer false ceiling inspections.</i> Primary responsibilities: • Conduct independent and collaborative research, contributing to larger projects under PI supervision. • Write research articles, proposals, and technical reports. • Mentored undergraduate and postgraduate students.
2021 – 2022	Visiting Postdoctoral Fellow in Laboratoire des Sciences du Numérique de Nantes (LS2N), École Centrale de Nantes (ECN). Supervisors: Prof. Stephane Caro and Prof. Darwin Lau, 26/10/2021 – 05/01/2022. <i>Project title: Iterative-learning control of cable-driven parallel robot.</i>
2017 – 2020	Graduate Teaching Assistant , The University of Auckland, New Zealand, 2017–2020.
2018 – 2019	Research Assistant at The University of Auckland, under the direction of Prof Peter Xu, 09/01/2018 – 08/01/2019.
2013 – 2016	Lecturer (Assistant Professor Grade-1) , School of Electrical, Electronics and Communication Engineering, Galgotias University, India, 19/07/2013–18/06/2016.

Teaching Experience

2026	DESE71002, <i>Robotics Research Projects</i>, Year 4/MSc (Design Engineering), Dyson School of Design Engineering, Imperial College London, United Kingdom. Guest lecture on <i>Robotics Simulation and Control using PyDrake</i> (28/01/2026).
2021 – 2024	MAEG5090, <i>Topics in Robotics</i>, for postgraduate students (MSc/MPhil), Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Hong Kong SAR. Class size: 50 students.
2017 – 2020	MECHENG 706, <i>Mechatronics Design Projects</i>, for 4th year undergraduate students, The University of Auckland, New Zealand.
2017 – 2020	MECHENG 306, <i>Design of Sensing and Actuating Systems</i>, for 3rd year undergraduate students, The University of Auckland, New Zealand.
2017 – 2020	ENGGEN 115, <i>Principles of Engineering Design</i>, for undergraduate students, The University of Auckland, New Zealand.
2017 – 2020	ENGGEN 131, <i>Introduction to Engineering Computation and Software Development</i>, for undergraduate students, The University of Auckland, New Zealand.

2013 – 2016	BECE2010 , <i>Digital Electronics</i> , for 2nd year undergraduate students (class size: 60), School of Electrical, Electronics and Communication Engineering, Galgotias University, India.
2013 – 2016	BECE2016 , <i>Signals and Systems</i> , for 2nd year undergraduate students (class size: 60), School of Electrical, Electronics and Communication Engineering, Galgotias University, India.
2013 – 2016	BTEE2002 , <i>Network Analysis and Synthesis</i> , for 2nd year undergraduate students (class size: 60), School of Electrical, Electronics and Communication Engineering, Galgotias University, India.
2013 – 2016	BEEE3002 , <i>Control Systems</i> , for 3rd year undergraduate students (class size: 60), School of Electrical, Electronics and Communication Engineering, Galgotias University, India.
2013 – 2016	BEEE2001 , <i>Electrical Measurement and Instrumentation</i> , for 2nd year undergraduate students (class size: 60), School of Electrical, Electronics and Communication Engineering, Galgotias University, India.
2013 – 2016	BECE3020 , <i>Digital Signal Processing</i> , for 3rd year undergraduate students (class size: 60), School of Electrical, Electronics and Communication Engineering, Galgotias University, India.
2013 – 2016	BTEE3019 , <i>Advanced Control System</i> , for 3rd year undergraduate students (class size: 60), School of Electrical, Electronics and Communication Engineering, Galgotias University, India.
2013 – 2016	BEEE5004 , <i>Digital Control</i> , for 1st year postgraduate students (class size: 20), School of Electrical, Electronics and Communication Engineering, Galgotias University, India.

Research Publications

Journal (Published/Accepted)

- [1] B. Jin*, A. Kobayashi*, D. Bhattacharya, A. Seino, F. Tokuda, N. Tien, and K. Kosuge. "Automated Straight-line Sewing of Stretchable Fabrics with Different Lengths". In: *IEEE Robotics and Automation Letters* (2025). *Equal contribution., pp. 1–8. DOI: [10.1109/LRA.2025.3619791](https://doi.org/10.1109/LRA.2025.3619791).
- [2] D. Bhattacharya, T. K. Cheung, Y. Wang, and D. Lau. "Kinematic and Dynamic Modeling of Cable-Object Interference and Wrapping in Complex Geometrical-Shaped Cable-Driven Parallel Robots". In: *Mechanism and Machine Theory* 214 (2025), p. 106092. DOI: [10.1016/j.mechmachtheory.2025.106092](https://doi.org/10.1016/j.mechmachtheory.2025.106092).
- [3] D. Bhattacharya, Y. P. Chan, S. Shang, Y. S. Chan, Y. Tan, and D. Lau. "Tri-Space Operational Control of Redundant Multilink and Hybrid Cable-Driven Parallel Robots Using an Iterative-Learning based Reactive Approach". In: *IEEE Transactions on Control Systems and Technology* (2023). DOI: [10.1109/TCST.2023.3263689](https://doi.org/10.1109/TCST.2023.3263689).
- [4] D. Bhattacharya, L. K. Cheng, and W. Xu. "Nonlinear Model Predictive Control of a Robotic Soft Esophagus for Endoprosthetic Stent Testing." In: *IEEE Transactions on Industrial Electronics* (2021). DOI: [10.1109/TIE.2021.3121755](https://doi.org/10.1109/TIE.2021.3121755).
- [5] D. Bhattacharya, S. Jesna, L. K. Cheng, and W. Xu. "RoSE: A Robotic Soft Esophagus for Endoprosthetic Stent Testing." In: *Soft Robotics* (2021). DOI: [10.1089/soro.2019.0205](https://doi.org/10.1089/soro.2019.0205).
- [6] Y. Dang, Y. Liu, R. Hashem, D. Bhattacharya, J. Allen, M. Stommel, L. K. Cheng, and W. Xu. "SoGut: A Soft Robotic Gastric Simulator". In: *Soft Robotics* (2021). DOI: [10.1089/soro.2019.0136](https://doi.org/10.1089/soro.2019.0136).
- [7] D. Bhattacharya, L. K. Cheng, and W. Xu. "Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot". In: *IEEE Transactions on Industrial Electronics* 67.6 (2020), pp. 4711–4720. DOI: [10.1109/TIE.2019.2928239](https://doi.org/10.1109/TIE.2019.2928239).

- [8] D. Bhattacharya, M. G. Nisha, and G. Pillai. "Relevance vector-machine-based solar cell model". In: *International Journal of Sustainable Energy* 34.10 (2015), pp. 685–692. DOI: [10.1080/14786451.2014.885030](https://doi.org/10.1080/14786451.2014.885030).

Books and Chapters

- [9] D. Bhattacharya, L. K. Cheng, S. Dirven, and W. Xu. *Artificial Intelligence Approach to the Trajectory Generation and Dynamics of a Soft Robotic Swallowing Simulator*. Cham: Springer International Publishing, 2019, pp. 3–16. DOI: [10.1007/978-3-319-78452-6_1](https://doi.org/10.1007/978-3-319-78452-6_1).

Conference Proceedings

- [10] W. Dong, D. Bhattacharya, A. Kobayashi, A. Seino, F. Tokuda, X. Huang, K. Tang, N. C. Tien, and K. Kosuge. "Precise Top-layer Fabric Segmentation for Fabric Destacking with Edge- and Shape-Aware Deep Networks". In: *2025 IEEE International Conference on Mechatronics and Automation (ICMA)*. 2025, pp. 1343–1348.
- [11] D. Bhattacharya, L. K. Cheng, and W. Xu. "Testing and Analysis of Migration Displacement of a Flared Stent Deployed in an Esophageal Swallowing Robot." In: *25th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*. 2018, pp. 1–6. DOI: [10.1109/M2VIP.2018.8600911](https://doi.org/10.1109/M2VIP.2018.8600911).
- [12] D. Bhattacharya, L. K. Cheng, S. Dirven, and W. Xu. "Actuation planning and modeling of a soft swallowing robot". In: *24th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*. 2017, pp. 1–6. DOI: [10.1109/M2VIP.2017.8211476](https://doi.org/10.1109/M2VIP.2017.8211476).

In Review/Preparation

- [13] D. Bhattacharya*, K. Tang*, X. Hang, F. Tokuda, N. Tien, and K. Kosuge. *RTFF: Random-to-Target Fabric Flattening Policy Using Dual-Arm Manipulator, IEEE International Conference on Robotics and Automation (ICRA)*. Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*. in review. *Equal contribution. 2026.
- [14] W. Wang, D. Bhattacharya, A. Kobayashi, A. Seino, F. Tokuda, N. Tien, and K. Kosuge. *Automated Fabric Alignment Using Global-Local Optimization and Point Cloud Registration, IEEE Transactions on Automation Science and Engineering (T-ASE)*. in review.
- [15] T. Cheung, D. Bhattacharya, and D. Lau. *Rapidly Deployable Deformable Frame Cable-Driven Parallel Robots with Small Footprint*. in preparation.
- [16] Y. Wang, D. Bhattacharya, and D. Lau. *Static Modelling and Workspace Analysis with Arbitrary Cable Attachment Design in a Cable-Driven Parallel Robot with Passive Backbone, IEEE Transactions on Automation Science and Engineering*. in preparation.

Grants and Scholarships

Grants/Fellowships

2025	Recipient, Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship for the project "CASREx" (Grant No. 101205678), hosted at Imperial College London . Funded under HORIZON-MSCA-2024-PF-01-01. Duration: 24 months, EUR 276,187.92 .
2023	Co-authored Innovation and Technology Fund (ITF), Partnership Research Program (PRP) grant.
2022	Co-authored Research Talent Hub for ITF project (RTH-ITF) grant.
2018	Recipient, Dynamic testing of implant stents using Robotic Soft Esophagus (RoSE), BvW Holding, Cham, Switzerland (39,000 NZD, 1 year) .

2018, 2019	Recipient, Riddet Travel grant, Riddet Institute Centre of Research Excellence (3,000 NZD).
Scholarships	
2019–2020	Recipient, Doctor of Philosophy Scholarship, University of Auckland.
2016–2019	Recipient, Doctor of Philosophy Scholarship, Riddet Institute Centre of Research Excellence.
2011–2013	Recipient, Masters of technology Scholarship, Ministry of Human Resource (MHRD) GATE scholarship.

Academic Awards, and Miscellaneous Achievements

Academic Awards

15/12/25	Selected, European Talent Academy 2026 (Imperial College London, TU Munich, Politecnico di Milano).
2017	Recipient, Best Conference paper award , 5th International Conference on Robotic Intelligence Technology and Applications (RiTA), Daejeon, Korea.
2017	Ranked 2nd in poster presentation in Delivery of Functionality in Complex Food Systems (DoF 2017) Conference.

Competitive Examinations

2011	Percentile: 99.12% (Rank 1,211/1,37,853) in Graduate Aptitude Test of Engineering (GATE).
2010	Percentile: 96.00% in Graduate Aptitude Test of Engineering (GATE).
2004	Ranked 31st in All India Junior mathematics Olympiad.

Professional Activities

05/02/26	Invited Speaker, Dyson Engineering Departmental Seminar , Imperial College London.
29/01/26	Invited Speaker, Imperial Robotics Postgrad Committee , Imperial College London. Talk: <i>Learning-based Fabric Manipulation with Mesh Perception and Visual Servoing</i> .
28/01/26	Delivered lecture on Robotics Simulation and Control using PyDrake for DESE71002 Robotics Research Projects, Dyson School of Engineering, Imperial College London.
08/08/25	Invited Speaker, World Robotics Conference (WRC 2025) , Beijing, China. Talk: <i>Fabric modeling and perception for garment manufacturing</i> .
03/08/25	Workshop Organizer and Presenter, IEEE ICMA 2025 , Beijing, China. Talk: <i>Mesh-based Fabric State Estimation and Manipulation</i> .
29/05/23	Poster Presentation, IEEE ICRA 2023 , ExCeL London, United Kingdom.
2018	Presenter, BioMouth 2018 Meet , Auckland, New Zealand.
2021 - Current	Reviewer of IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA), ASME Journal of Mechanisms and Robotics (JMR).
2017 - Current	Member of the Institute of Electrical and Electronics Engineers (IEEE).
2014	Former member of the Electrical Engineering Society (EES), IIT Roorkee, and NERIST Electronics Society (NES), NERIST.
2006	Have an 'Outstanding Certificate' in National Service Scheme (NSS) for contributing in ac-

tivities like blood donation camps, village cleanliness programmes and children education.

Skills

Languages	English and Hindi: professional reading, writing, and speaking proficiency.
Research areas	Soft robotics, cable-driven robots, deformable-object manipulation, robot modelling and control, visual servoing, learning-based control, and robotic systems for healthcare and manufacturing.
Programming	Python, MATLAB, PyTorch, Gym framework, ROS, LabVIEW, assembly-level programming (8085, 8086, MASM32, and 8051), Git, and LaTeX.
Software	Blender, SOLIDWORKS, Autodesk Inventor, Inkscape, and Lightroom.
Hardware	Raspberry Pi, myRIO, and Arduino.